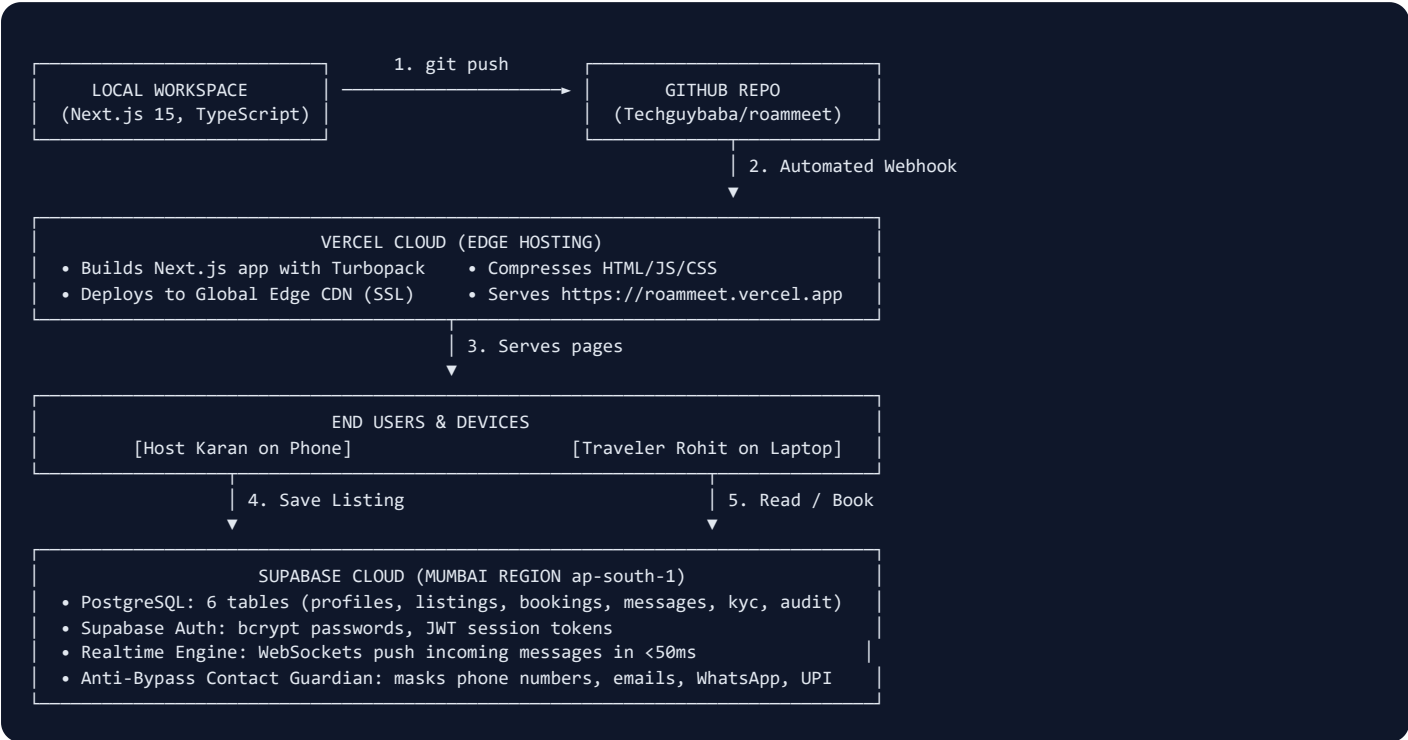


TECHNICAL ARCHITECTURE & FLOW GUIDE

RoamMeet: Platform Architecture & Data Flow

A complete breakdown of where code is developed, where it is stored, how Vercel deploys it, and how Supabase manages the cloud database & real-time messaging.



1 Local Code

Next.js 15, TypeScript, and Tailwind CSS. All code is tested and verified locally before shipping.

2 GitHub Cloud

Central Git repository on the `main` branch. Holds version history and triggers deployments.

3 Vercel Edge

Automated CI/CD build, Next.js optimization, global edge distribution, and automated SSL (HTTPS).

4 Supabase Mumbai

PostgreSQL database, real-time WebSockets, bcrypt auth, and Contact Shield running in Mumbai (ap-south-1).

⚡ What Happens on Vercel?

Vercel is the front-line hosting engine that makes the website accessible worldwide:

1. Automated Build Trigger

Whenever code is pushed to GitHub, Vercel starts an automated build container.

2. Compilation & Type Checking

Runs `next build --turbopack`. Validates TypeScript types and pre-renders static and dynamic routes.

3. Global Edge CDN Distribution

Deploys code across hundreds of global edge data centers for sub-second loading anywhere.

4. Automatic SSL & Security

Maintains HTTPS certificates and guards against DDoS attacks with zero downtime.

📄 What Happens on Supabase?

Supabase is the persistent cloud brain and database located in Mumbai (ap-south-1):

1. PostgreSQL Database Storage

Stores all user accounts, listings, bookings, chat messages, and KYC documents securely.

2. Authentication & Password Hashing

Uses industry-standard `bcrypt` encryption. Manages user sessions via JWT tokens.

3. Realtime WebSockets Engine

When a chat message or booking is submitted, Supabase broadcasts it live to connected devices in milliseconds.

4. Row Level Security (RLS)

Database-level security rules ensuring users only modify data they are authorized to touch.

 Complete Lifecycle of a Live User Action (Example)

Step 1: Host Publishes

Host Karan taps "Publish Listing" on his phone. The request is routed via Vercel to Supabase.

Step 2: Database Write

Supabase writes the record into the `listings` table in Mumbai and logs the \$1 / ₹79 fee rules.

Step 3: Traveler Discovers

Traveler Rohit visits the feed on his laptop. The site queries Supabase and displays the new stay instantly.

Step 4: Realtime Booking

Rohit books the stay. Both devices receive live WebSocket updates while private contacts stay locked.

Infrastructure Component	Provider	Region	Monthly Cost
Frontend & Global Edge CDN	Vercel	Global Edge Network	\$0.00 / month (Free Tier)
Cloud Database & Realtime	Supabase (PostgreSQL)	South Asia (Mumbai ap-south-1)	\$0.00 / month (Free Tier)
Source Control & CI/CD	GitHub	Global Cloud	\$0.00 / month (Free Tier)
SSL & Edge Encryption	Vercel / Let's Encrypt	Global	\$0.00 / month (Included)